CHAPTER 5

MAGNETISM AND MATTER

CHAPTER ESSENCE AT A GLANCE:
Earth's magnetic elements, bar magnet dipole analogies, Gauss's law for magnetism, and diamagnetic, paramagnetic, and ferromagnetic materials.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Magnetic Field Lines & Monopole Absence:** Magnetic lines form continuous closed loops with no beginning or end, confirming that isolated magnetic monopoles do not exist in nature.

• **Bar Magnet & Solenoid Equivalence:** A current-carrying solenoid produces an external magnetic field pattern identical to a magnetic dipole or bar magnet of equivalent moment.

• **Diamagnetism & Meissner Expulsion:** Diamagnetic materials develop induced moments opposing external fields, exhibiting weak repulsion, temperature independence, and complete field expulsion in superconductors.

• **Paramagnetism & Curie Alignment:** Paramagnetic substances possess permanent atomic magnetic dipoles that weakly align along applied fields, diminishing with increasing temperature as governed by Curie's law.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$B_{\text{axial}} = \frac{\mu_0}{4\pi}\frac{2M}{r^3}, \qquad B_{\text{eq}} = \frac{\mu_0}{4\pi}\frac{M}{r^3}$$

Magnetic field of a magnetic dipole (bar magnet)

$$\tau = \mathbf{M} \times \mathbf{B}$$

Torque on a magnetic dipole in a magnetic field

$$\chi = \frac{C}{T}$$

Curie's Law in paramagnetism

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** Continuous closed loops (unlike electric field lines); tangent gives field direction; never intersect.

• **Alert:** Superconductors exhibit perfect diamagnetism with χ = -1 and μ r = 0, completely expelling magnetic field lines.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Bar magnet as an equivalent solenoid [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Setup & Equivalence]** Solenoid of length 2l, radius a, n turns per unit length carrying current I. Axial field point P is at distance r from center (r >> a, r >> l). [1 Mark]

• **[Step 1: Axial Field Derivation]** Consider element of thickness dx at distance x from center containing n dx turns. dB = [μ_0 (n dx) I a^2] divided by [2 ((r - x)^2 + a^2)^(3 divided by 2)] ≈ [μ_0 n I a^2 dx] divided by [2 r^3] (since r >> x, a). B = $\int$ {-l}$^{+l}$ dB = [(μ_0 n I a^2) divided by (2 r^3)] $\int$ {-l}$^{+l}$ dx = [(μ_0 n I a^2) divided by (2 r^3)] · (2l) = [μ_0 I (n · 2l) (π a^2)] divided by (2π r^3). Total turns N = n · 2l, cross-section area A = π a^2, magnetic moment m = N I A · B = (μ_0 divided by 4π) · (2m divided by r^3). This is identical to the axial field of a bar magnet of dipole moment m! [2 Marks]

• **Key Exam Result:** Solenoid behaves as bar magnet with axial field B = (μ_0 divided by 4π) (2m divided by r^3)

2. Magnetic dipole moment of a revolving electron (Bohr magneton) [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Current & Magnetic Moment]** Electron of charge e, mass m e revolving in circular orbit of radius r with speed v and frequency ν = v divided by (2π r). Orbital current: I = e ν = e v divided by (2π r). Orbital magnetic dipole moment: μ l = I A = [e v divided by (2π r)] · (π r^2) = (half (0.5)) e v r. [1.5 Marks]

• **[Step 2: Linking to Angular Momentum & Bohr Magneton]** Orbital angular momentum L = m e v r ⇒ v r = L divided by m e. μ l = (e divided by 2 m e) L. Gyromagnetic ratio = μ l divided by L = e divided by (2 m e) = 8.8 × 10^{10} C/kg. By Bohr's quantization postulate: L = n h divided by (2π). For n = 1: μ B = e h divided by (4π m e) = (1.602 × 10^{-19} × 6.626 × 10^{-34}) divided by (4π × 9.109 × 10^{-31}) = 9.27 × 10^{-24} A m^2 (or $\frac{J}{T}$). This fundamental quantum is called the Bohr Magneton (μ B). [1.5 Marks]

• **Key Exam Result:** μ l = (e divided by 2 m e) L; Bohr Magneton μ B = e h divided by (4π m e) = 9.27 × 10^{-24} A m^2

3. Magnetic field lines and their key properties [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Key Properties of Magnetic Field Lines]** 1. Magnetic field lines form continuous closed loops (unlike electrostatic lines), emerging from North pole and entering South pole outside the magnet, and continuing from South to North inside. 2. The tangent to a field line at any point gives the direction of net magnetic field B at that point. 3. Field lines never intersect each other (intersection would mean two conflicting field directions at that point). 4. Line density is proportional to magnetic field strength (crowded near poles where field is strongest). [2 Marks]

• **Key Exam Result:** Field lines form continuous closed loops and never intersect

4. Gauss's law in magnetism: statement and physical significance [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Statement of Gauss's Law in Magnetism]** The net magnetic flux through any closed Gaussian surface S is identically zero: closed integral S B · dA = 0. [1 Mark]

• **[Physical Significance]** 1. Signifies that isolated magnetic poles (magnetic monopoles) DO NOT EXIST in nature. 2. Magnetic poles always exist in equal and opposite pairs (dipoles). Every field line entering a closed surface must also leave it, making net flux zero. [1 Mark]

• **Key Exam Result:** closed integral B · dA = 0; Proves non-existence of magnetic monopoles

5. Diamagnetic, paramagnetic, and ferromagnetic substances: comparison of properties [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Comparison Table: Dia, Para, and Ferromagnetic]** 1. Diamagnetic: Weakly repelled by magnets; move from stronger to weaker field. Susceptibility χ is small and negative ($-1 \leq \chi < 0$). Relative permeability $\mu_r < 1$. Temperature independent. Examples: Bismuth, Copper, Water. [1 Mark]

• 2. Paramagnetic: Weakly attracted by magnets; move from weaker to stronger field. Susceptibility χ is small and positive ($0 < \chi < \varepsilon$). Relative permeability $\mu_r > 1$. Obeys Curie's law (χ proportional to 1 divided by T). Examples: Aluminum, Oxygen, Platinum. [1 Mark]

• 3. Ferromagnetic: Strongly attracted by magnets; move strongly to strongest field regions. Form magnetic domains. Susceptibility χ is very large and positive ($\chi >> 1$). Relative permeability $\mu_r >> 1$. Above Curie temperature T_c, becomes paramagnetic. Examples: Iron, Cobalt, Nickel. [1 Mark]

• **Key Exam Result:** Dia: $\chi < 0$, repelled | Para: $\chi > 0$ small, attracted | Ferro: $\chi >> 1$, domains

6. Numerical on torque and work done on magnetic dipole [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Given Data & Formulas]** Magnetic moment m = 0.48 $\frac{J}{T}$, magnetic field B = 0.15 T. 1. Maximum torque: $\tau_{max} = m B \sin 90° = 0.48 \times 0.15 = 0.072$ N m. [1 Mark]

• **[Step 2: Work Done]** Work required to rotate from stable ($\theta_1 = 0°$) to unstable ($\theta_2 = 180°$): $W = - m B (\cos 180° - \cos 0°) = - m B (-1 - 1) = + 2 m B = 2 \times (0.48 \times 0.15) = 2 \times 0.072 = 0.144$ J. [1 Mark]

• **Key Exam Result:** Maximum torque $\tau_{max} = 0.072$ N m; Work done W = 0.144 J

7. Numerical on Bohr magneton calculation [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Formula & Constants]** Bohr Magneton $\mu_B = (e h)$ divided by $(4\pi m_e)$. $e = 1.602 \times 10^{-19}$ C, $h = 6.626 \times 10^{-34}$ J s, $m_e = 9.109 \times 10^{-31}$ kg. [1 Mark]

• **[Step 2: Calculation]** $\mu_B = (1.602 \times 10^{-19} \times 6.626 \times 10^{-34})$ divided by $(4 \times 3.1416 \times 9.109 \times 10^{-31}) = (1.0615 \times 10^{-52})$ divided by $(1.1447 \times 10^{-29}) = 9.273 \times 10^{-24}$ A m^2 (or $\frac{J}{T}$). [1 Mark]

• **Key Exam Result:** $\mu_B = 9.27 \times 10^{-24}$ A m^2